NPTEL Software Testing

100 NPTEL-Style Practice MCQs

Single-correct-answer practice set covering the concepts in your revision material. Answer key is only on the final page.

1. A defect in source code is called a:
A. failure
B. fault
C. oracle
D. test path

2. An incorrect internal program state is an:
A. error
B. edge
C. input block
D. mutant score

3. In RIPR, the final R means:
A. retest
B. revelation
C. restriction
D. recovery

4. A test executes a fault but state remains correct. Which RIPR condition failed?
A. Reachability
B. Infection
C. Propagation
D. Revelation

5. A weak assertion most directly harms:
A. reachability
B. infection
C. revelation
D. partitioning

6. Testing differs from debugging because debugging primarily:
A. generates partitions
B. locates and repairs faults after failure
C. measures load
D. covers edges

7. Coverage evidence alone proves:
A. absence of faults
B. all outputs correct
C. only that specified obligations executed
D. all mutants equivalent

8. The user-visible incorrect behavior is a:
A. definition
B. failure
C. clause
D. base choice

9. A test oracle determines:
A. which code compiles
B. pass or fail
C. graph nodes
D. runtime type

10. A propagated error not checked by the oracle fails at:
A. Reachability
B. Infection
C. Propagation
D. Revelation

11. For valid partitions, blocks must be:
A. random and equal-sized
B. complete and disjoint
C. numeric and ordered
D. all valid

12. Interface-based ISP mainly considers:
A. each parameter separately
B. only requirements
C. only invalid inputs
D. method calls

13. Functionality-based ISP mainly derives characteristics from:
A. file names
B. requirements and behavior
C. compiler output
D. CPU usage

14. For block counts 2,3,4, ACoC requires:
A. 4
B. 9
C. 12
D. 24

15. ECC requires:
A. every combination
B. every block at least once
C. every pair of paths
D. only base tests

16. Pair-wise coverage focuses on:
A. all 2-way block interactions
B. two test cases only
C. two values per input
D. all paths of length two

17. T-wise becomes ACoC when t equals:
A. 2
B. largest block count
C. number of characteristics
D. number of tests

18. BCC varies:
A. all characteristics simultaneously
B. one characteristic from a base at a time
C. only invalid blocks
D. only outputs

19. With blocks 3,2,4, BCC count is:
A. 7
B. 8
C. 9
D. 24

20. An impossible block combination is:
A. complete
B. infeasible
C. prime
D. equivalent

21. A test path must start at an initial node and:
A. repeat every node
B. end at a final node
C. contain a cycle
D. be prime

22. A prime path is:
A. any complete path
B. a maximal simple path
C. a path of one edge
D. an infeasible path

23. Which is stronger in the usual graph hierarchy?
A. Node Coverage over Edge Coverage
B. Edge Coverage over Node Coverage
C. Node and Edge always identical
D. Neither has requirements

24. Edge-Pair Coverage requires paths of length:
A. 0 edges
B. 1 edge
C. 2 edges
D. all edges

25. A direct tour means the required path appears:
A. in reverse
B. contiguously
C. only with a detour
D. in another method

26. A sidetrip typically:
A. leaves and returns while preserving required order
B. removes a node
C. changes the final node
D. makes a path simple

27. A continue statement in a loop transfers control to:
A. method exit
B. next loop condition/update
C. previous method
D. exception handler only

28. Complete Path Coverage may be impractical because loops can produce:
A. no paths
B. infinitely many test paths
C. one node
D. only prime paths

29. PPC stands for:
A. Predicate Pair Coverage
B. Prime Path Coverage
C. Program Partition Criterion
D. Pair Path Count

30. Edge Coverage can still miss:
A. all nodes
B. faults requiring particular edge sequences
C. compilation errors
D. every branch

31. A statement x=7 is a:
A. use
B. definition
C. predicate
D. final node

32. A use in y=x+1 is a:
A. p-use
B. c-use
C. definition
D. kill

33. A use in if(x>0) is a:
A. p-use
B. c-use only
C. definition
D. mutant

34. A def-clear path contains:
A. no use
B. no redefinition between def and use
C. no branches
D. no nodes

35. A du-pair connects:
A. two definitions
B. a definition and reachable use
C. two predicates
D. two mutants

36. The weakest of these is usually:
A. all-du-paths
B. all-uses
C. all-defs
D. combinatorial coverage

37. All-uses requires:
A. one use per program
B. every reachable use for each definition
C. every path in graph
D. all inputs

38. A redefinition between d and u:
A. preserves old def
B. kills the old reaching definition
C. creates PC
D. makes a prime path

39. Use before definition is a potential:
A. data-flow anomaly
B. base choice
C. load test
D. equivalent mutant

40. Coupling du-pairs are most relevant to:
A. interprocedural testing
B. font testing
C. ACoC
D. spike testing

41. An FSM transition is usually labelled by:
A. an event or condition
B. a block count
C. a mutation score
D. a file extension

42. Before testing a transition, a test may need a prefix to:
A. compile source
B. reach the pre-state
C. kill a mutant
D. form a clause

43. Transition coverage on an FSM is analogous to:
A. edge coverage
B. clause coverage
C. all-defs
D. ECC

44. Use-case scenarios are especially useful for:
A. user-visible flows
B. CPU arithmetic only
C. equivalent mutants
D. hash codes only

45. A call graph node commonly represents a:
A. method/function
B. input block
C. truth value
D. browser cookie

46. A predicate is an expression that returns:
A. a file
B. Boolean value
C. graph only
D. mutant

47. A clause contains:
A. no logical operator
B. at least two ANDs
C. a complete path
D. a definition and use

48. Predicate Coverage requires each predicate:
A. true only
B. false only
C. true and false
D. all 2^n combinations

49. Clause Coverage guarantees Predicate Coverage:
A. always
B. never under any circumstance
C. not in general
D. only for 10 clauses

50. For n independent clauses, CoC needs:
A. n
B. n+1
C. 2n
D. 2^n

51. For p=a OR b, a determines p when b is:
A. true
B. false
C. undefined
D. both true and false simultaneously

52. The clause currently being tested for influence is the:
A. minor clause
B. major clause
C. base clause
D. final clause

53. GACC permits minor-clause values to:
A. never change
B. differ between paired tests
C. be omitted from predicate
D. all be true only

54. CACC additionally requires the predicate outcomes to:
A. remain equal
B. differ
C. be infeasible
D. be ignored

55. RACC requires minor clauses in the pair to be:
A. identical
B. all true
C. all false
D. unreachable

56. Which criterion checks non-influence?
A. ACC
B. ICC
C. ACoC
D. EC

57. MC/DC is closely related to:
A. active clause coverage
B. load testing
C. all-defs
D. web caching

58. Constraints can make RACC:
A. automatic
B. infeasible
C. the same as ECC
D. a graph node

59. CACC subsumes:
A. Predicate Coverage
B. all-du-paths
C. ACoC
D. stress testing

60. GACC necessarily subsumes PC:
A. true
B. false
C. only for mobile apps
D. only for no clauses

61. A mutant is killed when:
A. it compiles
B. a test distinguishes it from original
C. it has no operator
D. it is equivalent

62. Mutation score excludes confirmed:
A. killed mutants
B. equivalent mutants
C. tests
D. operators

63. Replacing + with - is an:
A. arithmetic operator replacement
B. edge pair
C. base choice
D. load model

64. A live mutant may indicate:
A. weak tests or equivalence
B. perfect correctness
C. no source code
D. only a syntax error

65. Strong mutation requires error difference to:
A. exist immediately only
B. propagate to observable output
C. avoid reachability
D. be a partition

66. Weak mutation checks difference:
A. near mutated location
B. only at final output
C. in requirements only
D. after deployment

67. Equivalent mutants are difficult because they:
A. always crash
B. behave identically on all inputs
C. cannot compile
D. always change output

68. Statement deletion is a type of:
A. mutation operator
B. graph node
C. state machine
D. oracle

69. Integration mutation primarily targets:
A. calls and interactions
B. screen size only
C. fonts
D. page numbering

70. The fundamental aim of mutation is to assess:
A. test sensitivity to faults
B. developer rank
C. network bandwidth
D. document quality

71. Dynamic binding selects a method using the:
A. runtime receiver type
B. file name
C. declared variable name only
D. test count

72. An overriding fault occurs between:
A. superclass and subclass methods
B. two input blocks
C. two graph edges
D. two browsers

73. equals symmetry means:
A. x.equals(y) equals y.equals(x)
B. x equals null
C. all objects equal
D. hash codes differ

74. If two objects are equal, their hash codes must be:
A. different
B. equal
C. negative
D. prime

75. A yo-yo graph is associated with:
A. deep inheritance and overriding
B. pair-wise ISP
C. stress testing
D. web links

76. Polymorphism tests should vary:
A. runtime receiver types
B. only filenames
C. only test order
D. only screen color

77. Coupling coverage examines:
A. interactions across calls
B. only local constants
C. only UI text
D. page layout

78. Substitutability problems arise when a subtype:
A. violates expected supertype behavior
B. has a constructor
C. uses a variable
D. is compiled

79. Retest-all means:
A. rerun the whole test suite
B. run only failed tests
C. drop redundant tests
D. sort tests alphabetically

80. Regression test selection:
A. chooses affected tests
B. always runs all tests
C. changes requirements
D. measures load

81. Test minimization may risk losing:
A. fault-detection diversity
B. compilation
C. all source files
D. runtime types

82. Prioritization mainly:
A. reorders tests for earlier feedback
B. deletes all slow tests
C. changes outputs
D. creates mutants

83. A safe selection technique aims not to omit:
A. tests that could reveal change-related faults
B. all equivalent mutants
C. all users
D. all documents

84. A flaky test is one that:
A. passes and fails nondeterministically
B. always fails correctly
C. has no assertion
D. uses a base choice

85. Change impact analysis may use:
A. dependencies and call graphs
B. only font size
C. only page count
D. only clause truth tables

86. Load testing evaluates:
A. expected workload
B. only sudden spikes
C. only recovery
D. only invalid inputs

87. Stress testing pushes the system:
A. beyond expected capacity
B. only at zero load
C. through truth tables
D. through inheritance

88. Spike testing applies:
A. sudden workload changes
B. long stable workload only
C. mutation operators
D. input partitions only

89. Soak testing is designed for:
A. sustained execution and degradation
B. one request
C. source compilation
D. clause pairing

90. Authentication answers:
A. who are you
B. what may you do
C. how fast is it
D. which path is prime

91. Authorization answers:
A. what may you do
B. who are you
C. how many blocks
D. which mutant is equivalent

92. Recovery testing checks:
A. restart/failover and data integrity
B. only UI color
C. only pair-wise inputs
D. only node coverage

93. Interoperability testing focuses on:
A. components, protocols and formats working together
B. only local variables
C. only classes
D. only load

94. Client-side validation should be backed by:
A. server-side validation
B. no validation
C. only CSS
D. only caching

95. Hidden fields and cookies should be considered:
A. untrusted inputs
B. trusted because browser-generated
C. graph nodes only
D. equivalent mutants

96. Bypass testing sends requests:
A. without relying only on normal UI flow
B. only through screenshots
C. only while offline
D. only after mutation

97. A mobile emulator is best used for:
A. broad fast configuration coverage
B. all hardware truth
C. battery chemistry
D. final real-device certainty

98. Real mobile devices are important for:
A. hardware and platform-realistic behavior
B. truth tables only
C. prime paths only
D. ACoC formulas

99. Mobile lifecycle testing includes:
A. background, rotation and process death
B. only login
C. only compilation
D. only static pages

100. Permission testing should include:
A. denial, grant and revoke
B. grant only
C. no OS versions
D. only load

Answer Key

1 B	**2** A	**3** B	**4** B	**5** C	**6** B	**7** C	**8** B	**9** B	**10** D
11 B	**12** A	**13** B	**14** D	**15** B	**16** A	**17** C	**18** B	**19** A	**20** B
21 B	**22** B	**23** B	**24** C	**25** B	**26** A	**27** B	**28** B	**29** B	**30** B
31 B	**32** B	**33** A	**34** B	**35** B	**36** C	**37** B	**38** B	**39** A	**40** A
41 A	**42** B	**43** A	**44** A	**45** A	**46** B	**47** A	**48** C	**49** C	**50** D
51 B	**52** B	**53** B	**54** B	**55** A	**56** B	**57** A	**58** B	**59** A	**60** B
61 B	**62** B	**63** A	**64** A	**65** B	**66** A	**67** B	**68** A	**69** A	**70** A
71 A	**72** A	**73** A	**74** B	**75** A	**76** A	**77** A	**78** A	**79** A	**80** A
81 A	**82** A	**83** A	**84** A	**85** A	**86** A	**87** A	**88** A	**89** A	**90** A
91 A	**92** A	**93** A	**94** A	**95** A	**96** A	**97** A	**98** A	**99** A	**100** A